LAPORAN PRAKTIKUM MODUL 11, 12, DAN LATIHAN SOAL MATAKULIAH BASIS DATA

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KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI REPUBLIK INDONESIA

PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN KOMPUTER DAN BISNIS

POLITEKNIK NEGERI CILACAP

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LANGKAH-LANGKAH

1. Masukk pada direktori **mysql-bin**. Login pada direktori dengan **mysql -u <user> <password>.** Pada contoh user=**root** dan password = kosong

```
Administrator: Command Pro
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>cd c:/xampp/mysql/bin

c:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and oth
ers.

Type 'help;' or '\h' for help. Type '\c' to clear the current in
put statement.
```

2. Simpan dokumentasi perintah-perintah sql yang akan dijalankan,dengan semua perintah yang disimpan pada **drive d:** dan disimpan dalam **namafile.doc**

```
MariaDB [(none)]> \T d:\salwa\note\tgsmodul11.doc
Logging to file 'd:\salwa\note\tgsmodul11.doc'
```

MODUL 11

INNER JOIN DAN OUTER JOIN

```
#Tampil semua database yang dimiliki
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| balai_pengobatan   |
| db_akademik        |
| db_asetdesa        |
| db_datasiswa       |
| db_jadwal          |
| db_koleksibuku     |
| db_lagu            |
| db_optik           |
| db_perpustakaan    |
| db_pustaka         |
| db_univ            |
| information_schema |
| koperasi           |
| kuliah             |
| latmodul4          |
| lelang             |
| library            |
| management         |
```

```
| modul3             |
| modul4             |
| modul5             |
| modul7             |
| mysql              |
| neraca_percobaan   |
| nilai_siswa        |
| optik              |
| penjualan          |
| performance_schema |
| pethouse           |
| phpmyadmin         |
| playlist_edukasi   |
| puskesmas          |
| pustaka            |
| sekolah            |
| sikepend           |
| sipi4              |
| test               |
| toko_online        |
+--------------------+
38 rows in set (0.086 sec)

MariaDB [(none)]> #Pilih database yang akan digunakan, disini
menggunakan database kuliah;
MariaDB [(none)]> use kuliah;
Database changed
MariaDB [kuliah]> #INNER JOIN
MariaDB [kuliah]> #Menampilkan isi tabel Dosen melalui tabel Mahasiswa
MariaDB [kuliah]> SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a INNER JOIN dosen b ON a.id_dosen = b.id_dosen;
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| KURNIAWAN        | 12344    | 12344    | DWI NOVIA PRASETYANT |
| INDRA KUSUMA     | 12345    | 12345    | CAHYA VIKASARI.,M.En |
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| RIZAD RAHMAN     | 12344    | 12344    | DWI NOVIA PRASETYANT |
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| FITRIADI BUDIMAN | 12344    | 12344    | DWI NOVIA PRASETYANT |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
+------------------+----------+----------+----------------------+
8 rows in set (0.085 sec)

MariaDB [kuliah]> #Menampilkan isi tabel Dosen dan Mahasiswa dengan
LEFT JOIN
MariaDB [kuliah]> SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a LEFT JOIN dosen b ON a.id_dosen = b.id_dosen;
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| LALU HERMAN      | 12346    | NULL     | NULL                 |
| KURNIAWAN        | 12344    | 12344    | DWI NOVIA PRASETYANT |
| INDRA KUSUMA     | 12345    | 12345    | CAHYA VIKASARI.,M.En |
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| RIZAD RAHMAN     | 12344    | 12344    | DWI NOVIA PRASETYANT |
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
```

```
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| FITRIADI BUDIMAN | 12344    | 12344    | DWI NOVIA PRASETYANT |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
| HIDAYAT NUGRAHA  | 12346    | NULL     | NULL                 |
+------------------+----------+----------+----------------------+
10 rows in set (0.001 sec)

MariaDB [kuliah]> #Menampilkan isi tabel Dosen dan Mahasiswa dengan
RIGHT JOIN
MariaDB [kuliah]> SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a RIGHT JOIN dosen b ON a.id_dosen = b.id_dosen;
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| KURNIAWAN        | 12344    | 12344    | DWI NOVIA PRASETYANT |
| INDRA KUSUMA     | 12345    | 12345    | CAHYA VIKASARI.,M.En |
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| RIZAD RAHMAN     | 12344    | 12344    | DWI NOVIA PRASETYANT |
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| FITRIADI BUDIMAN | 12344    | 12344    | DWI NOVIA PRASETYANT |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
| NULL             | NULL     | 12342    | ANTONIUS AGUNG H.,M. |
+------------------+----------+----------+----------------------+
9 rows in set (0.001 sec)
```

MODUL 12

CROSS JOIN DAN UNION JOIN

```
MariaDB [kuliah]> #UNION JOIN
MariaDB [kuliah]> #Menampilkan tabel mahasiswa dan dosen dengan
menggunakan CROSS JOIN
MariaDB [kuliah]> SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a CROSS JOIN dosen b;
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| LALU HERMAN      | 12346    | 12341    | NUR WAHYU RAHADI.,M. |
| LALU HERMAN      | 12346    | 12342    | ANTONIUS AGUNG H.,M. |
| LALU HERMAN      | 12346    | 12343    | ANDESITA PRIHANTARA. |
| LALU HERMAN      | 12346    | 12344    | DWI NOVIA PRASETYANT |
| LALU HERMAN      | 12346    | 12345    | CAHYA VIKASARI.,M.En |
| KURNIAWAN        | 12344    | 12341    | NUR WAHYU RAHADI.,M. |
| KURNIAWAN        | 12344    | 12342    | ANTONIUS AGUNG H.,M. |
| KURNIAWAN        | 12344    | 12343    | ANDESITA PRIHANTARA. |
| KURNIAWAN        | 12344    | 12344    | DWI NOVIA PRASETYANT |
| KURNIAWAN        | 12344    | 12345    | CAHYA VIKASARI.,M.En |
| INDRA KUSUMA     | 12345    | 12341    | NUR WAHYU RAHADI.,M. |
| INDRA KUSUMA     | 12345    | 12342    | ANTONIUS AGUNG H.,M. |
| INDRA KUSUMA     | 12345    | 12343    | ANDESITA PRIHANTARA. |
| INDRA KUSUMA     | 12345    | 12344    | DWI NOVIA PRASETYANT |
| INDRA KUSUMA     | 12345    | 12345    | CAHYA VIKASARI.,M.En |
| KARMA MAULANA    | 12343    | 12341    | NUR WAHYU RAHADI.,M. |
| KARMA MAULANA    | 12343    | 12342    | ANTONIUS AGUNG H.,M. |
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| KARMA MAULANA    | 12343    | 12344    | DWI NOVIA PRASETYANT |
| KARMA MAULANA    | 12343    | 12345    | CAHYA VIKASARI.,M.En |
| RIZAD RAHMAN     | 12344    | 12341    | NUR WAHYU RAHADI.,M. |
```

```
| RIZAD RAHMAN     | 12344    | 12342    | ANTONIUS AGUNG H.,M. |
| RIZAD RAHMAN     | 12344    | 12343    | ANDESITA PRIHANTARA. |
| RIZAD RAHMAN     | 12344    | 12344    | DWI NOVIA PRASETYANT |
| RIZAD RAHMAN     | 12344    | 12345    | CAHYA VIKASARI.,M.En |
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| WAWAN ADI PUTRA  | 12341    | 12342    | ANTONIUS AGUNG H.,M. |
| WAWAN ADI PUTRA  | 12341    | 12343    | ANDESITA PRIHANTARA. |
| WAWAN ADI PUTRA  | 12341    | 12344    | DWI NOVIA PRASETYANT |
| WAWAN ADI PUTRA  | 12341    | 12345    | CAHYA VIKASARI.,M.En |
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| M TAUFIK HIDAYAT | 12341    | 12342    | ANTONIUS AGUNG H.,M. |
| M TAUFIK HIDAYAT | 12341    | 12343    | ANDESITA PRIHANTARA. |
| M TAUFIK HIDAYAT | 12341    | 12344    | DWI NOVIA PRASETYANT |
| M TAUFIK HIDAYAT | 12341    | 12345    | CAHYA VIKASARI.,M.En |
| FITRIADI BUDIMAN | 12344    | 12341    | NUR WAHYU RAHADI.,M. |
| FITRIADI BUDIMAN | 12344    | 12342    | ANTONIUS AGUNG H.,M. |
| FITRIADI BUDIMAN | 12344    | 12343    | ANDESITA PRIHANTARA. |
| FITRIADI BUDIMAN | 12344    | 12344    | DWI NOVIA PRASETYANT |
| FITRIADI BUDIMAN | 12344    | 12345    | CAHYA VIKASARI.,M.En |
| IDA KUSUMAWATI   | 12343    | 12341    | NUR WAHYU RAHADI.,M. |
| IDA KUSUMAWATI   | 12343    | 12342    | ANTONIUS AGUNG H.,M. |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
| IDA KUSUMAWATI   | 12343    | 12344    | DWI NOVIA PRASETYANT |
| IDA KUSUMAWATI   | 12343    | 12345    | CAHYA VIKASARI.,M.En |
| HIDAYAT NUGRAHA  | 12346    | 12341    | NUR WAHYU RAHADI.,M. |
| HIDAYAT NUGRAHA  | 12346    | 12342    | ANTONIUS AGUNG H.,M. |
| HIDAYAT NUGRAHA  | 12346    | 12343    | ANDESITA PRIHANTARA. |
| HIDAYAT NUGRAHA  | 12346    | 12344    | DWI NOVIA PRASETYANT |
| HIDAYAT NUGRAHA  | 12346    | 12345    | CAHYA VIKASARI.,M.En |
+------------------+----------+----------+----------------------+
50 rows in set (0.001 sec)

MariaDB [kuliah]> #Menampilkan tabel mahasiswa dan dosen dengan
menggunakan UNION JOIN
MariaDB [kuliah]> #CARA 1
MariaDB [kuliah]> SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a JOIN dosen b ON a.id_dosen = b.id_dosen WHERE
a.id_dosen='12341' OR a.id_dosen='12343';
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
+------------------+----------+----------+----------------------+
4 rows in set (0.001 sec)

MariaDB [kuliah]> #CARA2 UNION
MariaDB [kuliah]> (SELECT a.nama, a.id_dosen, b.id_dosen, b.nama FROM
mahasiswa a JOIN dosen b ON a.id_dosen = b.id_dosen WHERE
a.id_dosen='12341') UNION (SELECT a.nama, a.id_dosen, b.id_dosen,
b.nama FROM mahasiswa a JOIN dosen b ON a.id_dosen = b.id_dosen WHERE
a.id_dosen='12343');
+------------------+----------+----------+----------------------+
| nama             | id_dosen | id_dosen | nama                 |
+------------------+----------+----------+----------------------+
| WAWAN ADI PUTRA  | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
```

```
| M TAUFIK HIDAYAT | 12341    | 12341    | NUR WAHYU RAHADI.,M. |
| KARMA MAULANA    | 12343    | 12343    | ANDESITA PRIHANTARA. |
| IDA KUSUMAWATI   | 12343    | 12343    | ANDESITA PRIHANTARA. |
+------------------+----------+----------+----------------------+
4 rows in set (0.001 sec)

MariaDB [kuliah]> \c
```

LATIHAN SOAL

```
MariaDB [kuliah]> MariaDB [(none)]> #STUDI KASUS: RENTAL MOBIL DEF
#TUGAS 1
MariaDB [(none)]> #1. Buat database Rental Mobil DEF
MariaDB [(none)]> CREATE DATABASE rental;
Query OK, 1 row affected (0.001 sec)
MariaDB [(none)]> #Pilih database yang akan digunakan
MariaDB [(none)]> use rental;
Database changed
MariaDB [rental]> #2. Buat tabel sesuai dengan data yang terdapat pada
soal dengan atribut, tipe data, panjang field, dan key yang sudah
ditentukan
MariaDB [rental]> #Buat tabel pelanggan
MariaDB [rental]> CREATE TABLE pelanggan (
    -> id_pelanggan VARCHAR(5) NOT NULL PRIMARY KEY,
    -> nama VARCHAR(30) NOT NULL,
    ->   alamat VARCHAR(45),
    -> no_telp VARCHAR(15));
Query OK, 0 rows affected (0.018 sec)

MariaDB [rental]> #Tabel Mobil
MariaDB [rental]> CREATE TABLE mobil (
    -> id_mobil VARCHAR(7) NOT NULL PRIMARY KEY,
    -> merek VARCHAR(30) NOT NULL,
    -> tahun SMALLINT(4),
    -> harga_sewa DECIMAL(10,2) NOT NULL);
Query OK, 0 rows affected (0.023 sec)

MariaDB [rental]> #Tabel Rental
MariaDB [rental]> CREATE TABLE rental (
    ->   id_rental VARCHAR(5) NOT NULL PRIMARY KEY,
    ->   id_pelanggan VARCHAR(5) NOT NULL,
    ->   tgl_rental DATE NOT NULL);
Query OK, 0 rows affected (0.020 sec)

MariaDB [rental]> #Tabel Detail Rental
MariaDB [rental]> CREATE TABLE detail_rental (
    ->   id_rental VARCHAR(5) NOT NULL,
    ->   id_mobil VARCHAR(7) NOT NULL,
    ->   lama_rental SMALLINT(3) NOT NULL);
Query OK, 0 rows affected (0.021 sec)

MariaDB [rental]> #3. Menampilkan semua database yang ada
MariaDB [rental]> show databases;
+--------------------+
| Database           |
+--------------------+
| balai_pengobatan   |
| db_akademik        |
```

```
| db_asetdesa        |
| db_datasiswa       |
| db_jadwal          |
| db_koleksibuku     |
| db_lagu            |
| db_optik           |
| db_perpustakaan    |
| db_pustaka         |
| db_univ            |
| information_schema |
| koperasi           |
| kuliah             |
| latmodul4          |
| lelang             |
| library            |
| management         |
| modul3             |
| modul4             |
| modul5             |
| modul7             |
| mysql              |
| neraca_percobaan   |
| nilai_siswa        |
| optik              |
| penjualan          |
| performance_schema |
| pethouse           |
| phpmyadmin         |
| playlist_edukasi   |
| puskesmas          |
| pustaka            |
| rental             |
| sekolah            |
| sikepend           |
| sipi4              |
| test               |
| toko_online        |
+--------------------+
39 rows in set (0.001 sec)

MariaDB [rental]> #4. Menampilkan struktur tabel untuk semua tabel
MariaDB [rental]> #Tabel Pelanggan
MariaDB [rental]> desc pelanggan;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id_pelanggan | varchar(5)  | NO   | PRI | NULL    |       |
| nama         | varchar(30) | NO   |     | NULL    |       |
| alamat       | varchar(45) | YES  |     | NULL    |       |
| no_telp      | varchar(15) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
4 rows in set (0.029 sec)

MariaDB [rental]> #Tabel mobil
MariaDB [rental]> desc mobil;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
```

```
| id_mobil   | varchar(7)    | NO   | PRI | NULL    |       |
| merek      | varchar(30)   | NO   |     | NULL    |       |
| tahun      | smallint(4)   | YES  |     | NULL    |       |
| harga_sewa | decimal(10,2) | NO   |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
4 rows in set (0.016 sec)

MariaDB [rental]> #Tabel rental
MariaDB [rental]> desc rental;
+--------------+------------+------+-----+---------+-------+
| Field        | Type       | Null | Key | Default | Extra |
+--------------+------------+------+-----+---------+-------+
| id_rental    | varchar(5) | NO   | PRI | NULL    |       |
| id_pelanggan | varchar(5) | NO   |     | NULL    |       |
| tgl_rental   | date       | NO   |     | NULL    |       |
+--------------+------------+------+-----+---------+-------+
3 rows in set (0.021 sec)

MariaDB [rental]> #Tabel detail_rental
MariaDB [rental]> desc detail_rental;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| id_rental   | varchar(5)  | NO   |     | NULL    |       |
| id_mobil    | varchar(7)  | NO   |     | NULL    |       |
| lama_rental | smallint(3) | NO   |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.025 sec)

MariaDB [rental]> #5. Mengisikan tabel dengan data contoh yang telah
disediakan di atas menggunakan perintah INSERT
MariaDB [rental]> #INSERT tabel pelanggan
MariaDB [rental]> INSERT INTO pelanggan
(id_pelanggan,nama,alamat,no_telp) VALUES
    -> ('C001','Andi','Jl. Cempaka No.2','08123456'),
    -> ('C002','Budi','Jl. Melati No.15','08234567'),
    -> ('C003','Sari','Jl. Mawar No.10','08345678'),
    -> ('C004','Tini','Jl. Anggrek No.5','08456789');
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [rental]> #INSERT tabel mobil
MariaDB [rental]> INSERT INTO mobil (id_mobil,merek,tahun,harga_sewa)
VALUES
    -> ('M10001','Toyota Avanza',2020,300000),
    -> ('M10002','Honda Brio',2021,250000),
    -> ('M10003','Suzuki Ertiga',2019,280000),
    -> ('M10004','Daihatsu Xenia',2018,270000);
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [rental]> #INSERT tabel rental
MariaDB [rental]> INSERT INTO rental
(id_rental,id_pelanggan,tgl_rental) VALUES
    -> ('R001','C001','2023-06-01'),
    -> ('R002','C003','2023-06-03'),
    -> ('R003','C002','2023-06-05');
Query OK, 3 rows affected (0.011 sec)
```

```
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [rental]> #INSERT tabel detail_rental
MariaDB [rental]> INSERT INTO detail_rental
(id_rental,id_mobil,lama_rental) VALUES
    -> ('R001','M10001',3),
    -> ('R001','M10002',2),
    -> ('R002','M10003',1),
    -> ('R003','M10004',4);
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [rental]> #6. Menampilkan isi semua tabel
MariaDB [rental]> #Tabel pelanggan
MariaDB [rental]> SELECT * FROM pelanggan;
+--------------+------+------------------+----------+
| id_pelanggan | nama | alamat           | no_telp  |
+--------------+------+------------------+----------+
| C001         | Andi | Jl. Cempaka No.2 | 08123456 |
| C002         | Budi | Jl. Melati No.15 | 08234567 |
| C003         | Sari | Jl. Mawar No.10  | 08345678 |
| C004         | Tini | Jl. Anggrek No.5 | 08456789 |
+--------------+------+------------------+----------+
4 rows in set (0.000 sec)

MariaDB [rental]> #Tabel mobil
MariaDB [rental]> SELECT * FROM mobil;
+----------+----------------+-------+------------+
| id_mobil | merek          | tahun | harga_sewa |
+----------+----------------+-------+------------+
| M10001   | Toyota Avanza  |  2020 |  300000.00 |
| M10002   | Honda Brio     |  2021 |  250000.00 |
| M10003   | Suzuki Ertiga  |  2019 |  280000.00 |
| M10004   | Daihatsu Xenia |  2018 |  270000.00 |
+----------+----------------+-------+------------+
4 rows in set (0.000 sec)

MariaDB [rental]> #Tabel rental
MariaDB [rental]> SELECT * FROM rental;
+-----------+--------------+------------+
| id_rental | id_pelanggan | tgl_rental |
+-----------+--------------+------------+
| R001      | C001         | 2023-06-01 |
| R002      | C003         | 2023-06-03 |
| R003      | C002         | 2023-06-05 |
+-----------+--------------+------------+
3 rows in set (0.000 sec)

MariaDB [rental]> #Tabel detail_rental
MariaDB [rental]> SELECT * FROM detail_rental;
+-----------+----------+-------------+
| id_rental | id_mobil | lama_rental |
+-----------+----------+-------------+
| R001      | M10001   |           3 |
| R001      | M10002   |           2 |
| R002      | M10003   |           1 |
| R003      | M10004   |           4 |
+-----------+----------+-------------+
```

```
4 rows in set (0.000 sec)
```

#TUGAS 2

```
MariaDB [rental]> #7. Menampilkan data semua mobil yang diproduksi pada
tahun 2020 atau lebih baru
MariaDB [rental]> SELECT * FROM mobil WHERE tahun >= 2020;
+----------+---------------+-------+------------+
| id_mobil | merek         | tahun | harga_sewa |
+----------+---------------+-------+------------+
| M10001   | Toyota Avanza |  2020 |  300000.00 |
| M10002   | Honda Brio    |  2021 |  250000.00 |
+----------+---------------+-------+------------+
2 rows in set (0.001 sec)

MariaDB [rental]> #8. Tampilkan semua nama pelanggan yang dimulai
dengan huruf "B"
MariaDB [rental]> SELECT * FROM pelanggan WHERE nama LIKE 'B%';
+--------------+------+------------------+----------+
| id_pelanggan | nama | alamat           | no_telp  |
+--------------+------+------------------+----------+
| C002         | Budi | Jl. Melati No.15 | 08234567 |
+--------------+------+------------------+----------+
1 row in set (0.000 sec)

MariaDB [rental]> #9. Tampilkan semua nama pelanggan yang tidak
diakhiri dengan huruf "i"
MariaDB [rental]> SELECT * FROM pelanggan WHERE nama NOT LIKE '%i';
Empty set (0.000 sec)

MariaDB [rental]> #10. Tampilkan 2 data pelanggan mulai dari data ke-3
MariaDB [rental]> SELECT * FROM pelanggan ORDER BY id_pelanggan LIMIT 2
OFFSET 2;
+--------------+------+------------------+----------+
| id_pelanggan | nama | alamat           | no_telp  |
+--------------+------+------------------+----------+
| C003         | Sari | Jl. Mawar No.10  | 08345678 |
| C004         | Tini | Jl. Anggrek No.5 | 08456789 |
+--------------+------+------------------+----------+
2 rows in set (0.000 sec)

MariaDB [rental]> #11. Hitung jumlah mobil berdasarkan merek
MariaDB [rental]> SELECT merek, COUNT(*) AS jumlah_mobil FROM mobil
GROUP BY merek;
+----------------+--------------+
| merek          | jumlah_mobil |
+----------------+--------------+
| Daihatsu Xenia |            1 |
| Honda Brio     |            1 |
| Suzuki Ertiga  |            1 |
| Toyota Avanza  |            1 |
+----------------+--------------+
4 rows in set (0.001 sec)

MariaDB [rental]> #12. Urutkan data pelanggan berdasarkan nama
pelanggan secara ascending
MariaDB [rental]> SELECT * FROM pelanggan ORDER BY nama ASC;
+--------------+------+------------------+----------+
| id_pelanggan | nama | alamat           | no_telp  |
```

```
+--------------+------+------------------+----------+
| C001         | Andi | Jl. Cempaka No.2 | 08123456 |
| C002         | Budi | Jl. Melati No.15 | 08234567 |
| C003         | Sari | Jl. Mawar No.10  | 08345678 |
| C004         | Tini | Jl. Anggrek No.5 | 08456789 |
+--------------+------+------------------+----------+
4 rows in set (0.000 sec)
```

MariaDB [rental]> **#13. Tampilkan data mobil yang diproduksi antara tahun 2018 hingga 2021**

```
MariaDB [rental]> SELECT * FROM mobil WHERE tahun BETWEEN 2018 AND
2021;
+----------+----------------+-------+------------+
| id_mobil | merek          | tahun | harga_sewa |
+----------+----------------+-------+------------+
| M10001   | Toyota Avanza  |  2020 |  300000.00 |
| M10002   | Honda Brio     |  2021 |  250000.00 |
| M10003   | Suzuki Ertiga  |  2019 |  280000.00 |
| M10004   | Daihatsu Xenia |  2018 |  270000.00 |
+----------+----------------+-------+------------+
4 rows in set (0.000 sec)
```

#TUGAS 3

MariaDB [rental]> **#14. Hitung total lama rental untuk semua mobil**

```
MariaDB [rental]> SELECT SUM(lama_rental) AS total_lama_rental FROM
detail_rental;
+-------------------+
| total_lama_rental |
+-------------------+
|                10 |
+-------------------+
1 row in set (0.000 sec)
```

MariaDB [rental]> **#15. Hitung rata-rata lama rental mobil**

```
MariaDB [rental]> SELECT AVG(lama_rental) AS rata_rata_lama FROM
detail_rental;
+----------------+
| rata_rata_lama |
+----------------+
|         2.5000 |
+----------------+
1 row in set (0.000 sec)
```

MariaDB [rental]> **#16. Tampilkan data mobil yang termurah**

```
MariaDB [rental]> SELECT * FROM mobil ORDER BY harga_sewa ASC LIMIT 1;
+----------+------------+-------+------------+
| id_mobil | merek      | tahun | harga_sewa |
+----------+------------+-------+------------+
| M10002   | Honda Brio |  2021 |  250000.00 |
+----------+------------+-------+------------+
1 row in set (0.000 sec)
```

MariaDB [rental]> **#17. Tampilkan data mobil yang termahal**

```
MariaDB [rental]> SELECT * FROM mobil ORDER BY harga_sewa DESC LIMIT 1;
+----------+---------------+-------+------------+
| id_mobil | merek         | tahun | harga_sewa |
+----------+---------------+-------+------------+
| M10001   | Toyota Avanza |  2020 |  300000.00 |
```

```
+----------+---------------+-------+------------+
1 row in set (0.000 sec)
```

#TUGAS 4 (TANPA MENGGUNAKAN JOIN)

```
MariaDB [rental]> #18. Tampilkan nama, alamat, dan nomor telepon
pelanggan yang merental mobil antara tanggal 2023-06-01 sampai dengan
2023-06-05
MariaDB [rental]> SELECT nama, alamat, no_telp FROM pelanggan WHERE
id_pelanggan IN (
    -> SELECT id_pelanggan FROM rental
    -> WHERE tgl_rental BETWEEN '2023-06-01' AND '2023-06-05'
    -> );
+------+------------------+----------+
| nama | alamat           | no_telp  |
+------+------------------+----------+
| Andi | Jl. Cempaka No.2 | 08123456 |
| Budi | Jl. Melati No.15 | 08234567 |
| Sari | Jl. Mawar No.10  | 08345678 |
+------+------------------+----------+
3 rows in set (0.001 sec)

MariaDB [rental]> #19. Tampilkan data pelanggan yang memiliki id_rental
"R002"
MariaDB [rental]> SELECT * FROM pelanggan
    -> WHERE id_pelanggan = (
    ->   SELECT id_pelanggan FROM rental WHERE id_rental = 'R002'
    -> );
+--------------+------+-----------------+----------+
| id_pelanggan | nama | alamat          | no_telp  |
+--------------+------+-----------------+----------+
| C003         | Sari | Jl. Mawar No.10 | 08345678 |
+--------------+------+-----------------+----------+
1 row in set (0.000 sec)

MariaDB [rental]> #20. Tampilkan merek mobil, tahun, dan harga sewa
yang dirental oleh pelanggan dengan id_rental "R001"
MariaDB [rental]> SELECT merek, tahun, harga_sewa
    -> FROM mobil
    -> WHERE id_mobil IN (
    ->   SELECT id_mobil FROM detail_rental WHERE id_rental = 'R001'
    -> );
+---------------+-------+------------+
| merek         | tahun | harga_sewa |
+---------------+-------+------------+
| Toyota Avanza |  2020 |  300000.00 |
| Honda Brio    |  2021 |  250000.00 |
+---------------+-------+------------+
2 rows in set (0.000 sec)

MariaDB [rental]> #21. Tampilkan data mobil yang dirental pada tanggal
2023-06-05
MariaDB [rental]> SELECT * FROM mobil
    -> WHERE id_mobil IN (
    -> SELECT id_mobil FROM detail_rental
    -> WHERE id_rental IN (
    -> SELECT id_rental FROM rental WHERE tgl_rental = '2023-06-05'));
+----------+---------------+-------+------------+
| id_mobil | merek         | tahun | harga_sewa |
```

```
+----------+----------------+-------+------------+
| M10004   | Daihatsu Xenia |  2018 |  270000.00 |
+----------+----------------+-------+------------+
1 row in set (0.000 sec)
```

MariaDB [rental]> **#22. Hitung jumlah mobil yang diproduksi oleh Toyota**

```
MariaDB [rental]> SELECT COUNT(*) AS jumlah_toyota FROM mobil WHERE
merek LIKE 'Toyota%';
+---------------+
| jumlah_toyota |
+---------------+
|             1 |
+---------------+
1 row in set (0.000 sec)
```

#TUGAS 5 (MENGGUNAKAN JOIN)

MariaDB [rental]> **#23. Tampilkan nama penyewa, alamat, nomor telepon, dan tanggal rental dengan menggunakan INNER JOIN**

```
MariaDB [rental]> SELECT p.nama, p.alamat, p.no_telp, r.tgl_rental FROM
pelanggan p INNER JOIN rental r ON p.id_pelanggan = r.id_pelanggan;
+------+------------------+----------+------------+
| nama | alamat           | no_telp  | tgl_rental |
+------+------------------+----------+------------+
| Andi | Jl. Cempaka No.2 | 08123456 | 2023-06-01 |
| Budi | Jl. Melati No.15 | 08234567 | 2023-06-05 |
| Sari | Jl. Mawar No.10  | 08345678 | 2023-06-03 |
+------+------------------+----------+------------+
3 rows in set (0.001 sec)
```

MariaDB [rental]> **#24. Tampilkan nama penyewa, alamat, nomor telepon, dan tanggal rental dengan menggunakan LEFT JOIN**

```
MariaDB [rental]> SELECT p.nama, p.alamat, p.no_telp, r.tgl_rental FROM
pelanggan p LEFT JOIN rental r ON p.id_pelanggan = r.id_pelanggan;
+------+------------------+----------+------------+
| nama | alamat           | no_telp  | tgl_rental |
+------+------------------+----------+------------+
| Andi | Jl. Cempaka No.2 | 08123456 | 2023-06-01 |
| Sari | Jl. Mawar No.10  | 08345678 | 2023-06-03 |
| Budi | Jl. Melati No.15 | 08234567 | 2023-06-05 |
| Tini | Jl. Anggrek No.5 | 08456789 | NULL       |
+------+------------------+----------+------------+
4 rows in set (0.000 sec)
```

MariaDB [rental]> **#25. Tampilkan nama penyewa, alamat, nomor telepon, dan tanggal rental dengan menggunakan RIGHT JOIN**

```
MariaDB [rental]> SELECT p.nama, p.alamat, p.no_telp, r.tgl_rental FROM
pelanggan p RIGHT JOIN rental r ON p.id_pelanggan = r.id_pelanggan;
+------+------------------+----------+------------+
| nama | alamat           | no_telp  | tgl_rental |
+------+------------------+----------+------------+
| Andi | Jl. Cempaka No.2 | 08123456 | 2023-06-01 |
| Sari | Jl. Mawar No.10  | 08345678 | 2023-06-03 |
| Budi | Jl. Melati No.15 | 08234567 | 2023-06-05 |
+------+------------------+----------+------------+
3 rows in set (0.001 sec)
```

MariaDB [rental]> **#26. Tampilkan data pelanggan dan rental menggunakan CROSS JOIN**

```
MariaDB [rental]> SELECT p.id_pelanggan, p.nama AS pelanggan,
r.id_rental, r.tgl_rental FROM pelanggan p CROSS JOIN rental r;
+--------------+-----------+-----------+------------+
| id_pelanggan | pelanggan | id_rental | tgl_rental |
+--------------+-----------+-----------+------------+
| C001         | Andi      | R001      | 2023-06-01 |
| C001         | Andi      | R002      | 2023-06-03 |
| C001         | Andi      | R003      | 2023-06-05 |
| C002         | Budi      | R001      | 2023-06-01 |
| C002         | Budi      | R002      | 2023-06-03 |
| C002         | Budi      | R003      | 2023-06-05 |
| C003         | Sari      | R001      | 2023-06-01 |
| C003         | Sari      | R002      | 2023-06-03 |
| C003         | Sari      | R003      | 2023-06-05 |
| C004         | Tini      | R001      | 2023-06-01 |
| C004         | Tini      | R002      | 2023-06-03 |
| C004         | Tini      | R003      | 2023-06-05 |
+--------------+-----------+-----------+------------+
12 rows in set (0.000 sec)
```

MariaDB [rental]> **#27. Tampilkan data pelanggan dan rental menggunakan UNION dengan id_pelanggan = 'C002' atau id_pelanggan = 'C004'**

```
MariaDB [rental]> (SELECT p.id_pelanggan, p.nama, r.id_rental  FROM
pelanggan p JOIN rental r ON p.id_pelanggan = r.id_pelanggan WHERE
p.id_pelanggan = 'C002')
    -> UNION
    -> (SELECT p.id_pelanggan, p.nama, r.id_rental FROM pelanggan p
JOIN rental r ON p.id_pelanggan = r.id_pelanggan WHERE p.id_pelanggan =
'C004');
+--------------+------+-----------+
| id_pelanggan | nama | id_rental |
+--------------+------+-----------+
| C002         | Budi | R003      |
+--------------+------+-----------+
1 row in set (0.001 sec)
```

#TUGAS 6

MariaDB [rental]> **#28. Hapus semua data dalam semua tabel, lalu tampilkan isi semua tabel**

```
MariaDB [rental]> TRUNCATE TABLE detail_rental;
Query OK, 0 rows affected (0.025 sec)

MariaDB [rental]> TRUNCATE TABLE rental;
Query OK, 0 rows affected (0.025 sec)

MariaDB [rental]> TRUNCATE TABLE mobil;
Query OK, 0 rows affected (0.022 sec)

MariaDB [rental]> TRUNCATE TABLE pelanggan;
Query OK, 0 rows affected (0.024 sec)
```

MariaDB [rental]> **#Menampilkan isi semua tabel**

MariaDB [rental]> **#Tabel detail_rental**

```
MariaDB [rental]> SELECT * FROM detail_rental;
Empty set (0.000 sec)
```

MariaDB [rental]> **#Tabel rental**

```
MariaDB [rental]> SELECT * FROM rental;
```

```
Empty set (0.000 sec)

MariaDB [rental]> #Tabel mobil
MariaDB [rental]> SELECT * FROM mobil;
Empty set (0.000 sec)

MariaDB [rental]> #Tabel pelanggan
MariaDB [rental]> SELECT * FROM pelanggan;
Empty set (0.000 sec)

MariaDB [rental]> #29. Hapus semua tabel, lalu tampilkan tabel dalam
database Rental Mobil DEF
MariaDB [rental]> DROP TABLE detail_rental;
Query OK, 0 rows affected (0.017 sec)

MariaDB [rental]> DROP TABLE rental;
Query OK, 0 rows affected (0.018 sec)

MariaDB [rental]> DROP TABLE mobil;
Query OK, 0 rows affected (0.007 sec)

MariaDB [rental]> DROP TABLE pelanggan;
Query OK, 0 rows affected (0.018 sec)

MariaDB [rental]> #30. Hapus database Rental Mobil DEF, lalu tampilkan
semua database yang ada
MariaDB [rental]> DROP DATABASE rental;
Query OK, 0 rows affected (0.024 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| balai_pengobatan   |
| db_akademik        |
| db_asetdesa        |
| db_datasiswa       |
| db_jadwal          |
| db_koleksibuku     |
| db_lagu            |
| db_optik           |
| db_perpustakaan    |
| db_pustaka         |
| db_univ            |
| information_schema |
| koperasi           |
| kuliah             |
| latmodul4          |
| lelang             |
| library            |
| management         |
| modul3             |
| modul4             |
| modul5             |
| modul7             |
| mysql              |
| neraca_percobaan   |
| nilai_siswa        |
```

```
| optik              |
| penjualan          |
| performance_schema |
| pethouse           |
| phpmyadmin         |
| playlist_edukasi   |
| puskesmas          |
| pustaka            |
| sekolah            |
| sikepend           |
| sipi4              |
| test               |
| toko_online        |
+--------------------+
38 rows in set (0.001 sec)
```